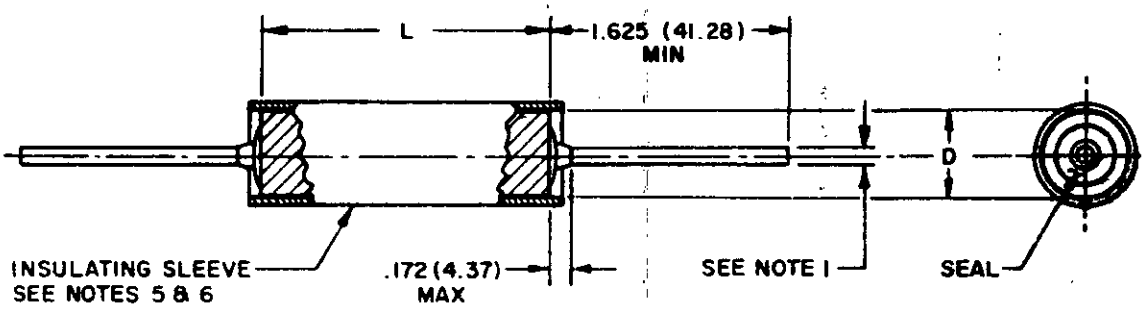
MILITARY SPECIFICATION SHEET

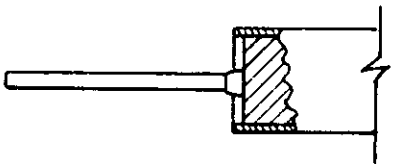
Ⓒ CAPACITORS, FIXED, METALLIZED, PAPER-PLASTIC FILM DIELECTRIC, DIRECT CURRENT. (HERMETICALLY SEALED IN METAL CASES). ESTABLISHED RELIABILITY. STYLE CHR19 (INSULATED)

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-39022.

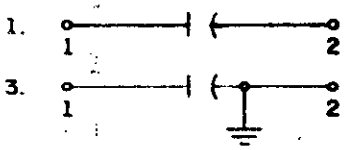
This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



CASE FOR CIRCUIT 1



CASE FOR CIRCUIT 3



CIRCUIT DIAGRAM

NOTES:
1. Number 22 AWG wire for case diameters of .312 (7.92 mm). Number 20 AWG wire for case diameters over .312 (7.92 mm).
2. See table I for additional dimensions.
3. Dimensions are in inches.
4. Metric equivalents (to the nearest .01 mm) are given for general information only, and are based upon 1 inch = 25.4 mm.
5. Insulating sleeve shall extend beyond the capacitor body but shall not exceed .031 (.79 mm) inch on either end. Insulating sleeve thickness shall not exceed .016 (.41 mm) inch.
6. Plastic insulating sleeve shall be transparent; marking shall be applied to the capacitor case.
7. Metric equivalents are in parentheses.

Ⓒ denotes changes.

FSC 5910

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:13Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figure and table I.

Case material: Nonmagnetic.

Capacitance value: See table I.

Capacitance tolerance: ± 5 percent and ± 10 percent.

Rated voltage: See table I.

Dielectric material: Normally paper-polyethylene-terephthalate.

Operating temperature range: -55° to +125°C.

Failure rate level: L (5.0%), M (1.0%), P (0.1%), R (0.01%), and S (0.001%).

Radiographic inspection (X-ray): Method 209 of MIL-STD-202.

Temperature cycling: Method 102 of MIL-STD-202, condition D, except step 3 shall be +125°+4, -0°C.

Seal: Method 112 of MIL-STD-202, condition A, or (for liquid-filled units only) 1 hour at +125°+4, -0°C with no evidence of leakage.

Dielectric withstanding voltage (DWV): Method 301 of MIL-STD-202.
Terminal to terminal - 200 percent of rated voltage.
Terminals to case (when case is not a terminal) - 200 percent of rated voltage.

NOTE: For 100-percent inspection:
250 percent for not less than 5 seconds, or
200 percent for not less than 15 seconds.

Insulation resistance (IR): Method 302 of MIL-STD-202.
Terminal to terminal - See table III.
Terminals to case (when case is not a terminal) - 10,000 megohms, minimum.

Capacitance (Cap.): Method 305 of MIL-STD-202.

Dissipation factor (DF): 1 percent.

Barometric pressure (reduced): Method 105 of MIL-STD-202, condition D (100,000 feet).
125 percent of rated voltage applied. See MIL-C-39022 for voltage limitations.

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20G).
50 percent of rated voltage applied.

Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hours).
Salt solution - 5 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:13Z
Check the source to verify that this is the current version before use.

© TABLE I. Electrical characteristics, dimensions and dash numbers.

DC rated voltage	Capacitance value	Circuit diagram	Capacitance tolerance	Dimensions, nominal (inches) 1/ 2/		Dash number Failure rate level for (%/1, 000 hr)				
volts	µF		Percent	L ±0.062	D ±0.031	L (5.0)	M (1.0)	P (0.1)	R (0.01)	S (0.001)
200	0.10	1	5	1.125	0.312	1001	1111	1221	1331	1441
200	0.10	1	10	1.125	0.312	1002	1112	1222	1332	1442
200	0.10	3	5	1.062	0.312	1003	1113	1223	1333	1443
200	0.10	3	10	1.062	0.312	1004	1114	1224	1334	1444
200	0.15	1	5	1.125	0.312	1005	1115	1225	1335	1445
200	0.15	1	10	1.125	0.312	1006	1116	1226	1336	1446
200	0.15	3	5	1.062	0.312	1007	1117	1227	1337	1447
200	0.15	3	10	1.062	0.312	1008	1118	1228	1338	1448
200	0.22	1	5	1.125	0.400	1009	1119	1229	1339	1449
200	0.22	1	10	1.125	0.400	1010	1120	1230	1340	1450
200	0.22	3	5	1.062	0.400	1011	1121	1231	1341	1451
200	0.22	3	10	1.062	0.400	1012	1122	1232	1342	1452
200	0.33	1	5	1.125	0.500	1013	1123	1233	1343	1453
200	0.33	1	10	1.125	0.500	1014	1124	1234	1344	1454
200	0.33	3	5	1.062	0.500	1015	1125	1235	1345	1455
200	0.33	3	10	1.062	0.500	1016	1126	1236	1346	1456
200	0.47	1	5	1.125	0.500	1017	1127	1237	1347	1457
200	0.47	1	10	1.125	0.500	1018	1128	1238	1348	1458
200	0.47	3	5	1.062	0.500	1019	1129	1239	1349	1459
200	0.47	3	10	1.062	0.500	1020	1130	1240	1350	1460
200	0.68	1	5	1.375	0.562	1021	1131	1241	1351	1461
200	0.68	1	10	1.375	0.562	1022	1132	1242	1352	1462
200	0.68	3	5	1.312	0.562	1023	1133	1243	1353	1463
200	0.68	3	10	1.312	0.562	1024	1134	1244	1354	1464
200	1.0	1	5	1.625	0.562	1025	1135	1245	1355	1465
200	1.0	1	10	1.625	0.562	1026	1136	1246	1356	1466
200	1.0	3	5	1.562	0.562	1027	1137	1247	1357	1467
200	1.0	3	10	1.562	0.562	1028	1138	1248	1358	1468
200	1.5	1	5	1.844	0.562	1029	1139	1249	1359	1469
200	1.5	1	10	1.844	0.562	1030	1140	1250	1360	1470
200	1.5	3	5	1.781	0.562	1031	1141	1251	1361	1471
200	1.5	3	10	1.781	0.562	1032	1142	1252	1362	1472
200	2.0	1	5	1.844	0.672	1033	1143	1253	1363	1473
200	2.0	1	10	1.844	0.672	1034	1144	1254	1364	1474
200	2.0	3	5	1.781	0.672	1035	1145	1255	1365	1475
200	2.0	3	10	1.781	0.672	1036	1146	1256	1366	1476
200	3.0	1	5	2.125	0.750	1037	1147	1257	1367	1477
200	3.0	1	10	2.125	0.750	1038	1148	1258	1368	1478
200	3.0	3	5	2.062	0.750	1039	1149	1259	1369	1479
200	3.0	3	10	2.062	0.750	1040	1150	1260	1370	1480
200	4.3	1	5	2.625	0.750	1041	1151	1261	1371	1481
200	4.3	1	10	2.625	0.750	1042	1152	1262	1372	1482
200	4.3	3	5	2.562	0.750	1043	1153	1263	1373	1483
200	4.3	3	10	2.562	0.750	1044	1154	1264	1374	1484
200	5.1	1	5	1.875	1.000	1045	1155	1265	1375	1485
200	5.1	1	10	1.875	1.000	1046	1156	1266	1376	1486
200	5.1	3	5	1.812	1.000	1047	1157	1267	1377	1487
200	5.1	3	10	1.812	1.000	1048	1158	1268	1378	1488
200	6.2	1	5	2.125	1.000	1049	1159	1269	1379	1489
200	6.2	1	10	2.125	1.000	1050	1160	1270	1380	1490
200	6.2	3	5	2.062	1.000	1051	1161	1271	1381	1491
200	6.2	3	10	2.062	1.000	1052	1162	1272	1382	1492

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:13Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical characteristics, dimensions and dash numbers - Continued.

DC rated voltage	Capacitance value	Circuit diagram	Capacitance tolerance	Dimensions, nominal (inches) 1/ 2/		Dash number Failure rate level for (%/1,000 hr)				
volts	µF		Percent	L ±0.062	D ±0.031	L (5.0)	M (1.0)	P (0.1)	R (0.01)	S (0.001)
200	8.2	1	5	2.625	1.000	1053	1163	1273	1383	1493
200	8.2	1	10	2.625	1.000	1054	1164	1274	1384	1494
200	8.2	3	5	2.562	1.000	1055	1165	1275	1385	1495
200	8.2	3	10	2.562	1.000	1056	1166	1276	1386	1496
400	0.047	1	5	1.125	0.400	1057	1167	1277	1387	1497
400	0.047	1	10	1.125	0.400	1058	1168	1278	1388	1498
400	0.047	3	5	1.062	0.400	1059	1169	1279	1389	1499
400	0.047	3	10	1.062	0.400	1060	1170	1280	1390	1500
400	0.068	1	5	1.125	0.400	1061	1171	1281	1391	1501
400	0.068	1	10	1.125	0.400	1062	1172	1282	1392	1502
400	0.068	3	5	1.062	0.400	1063	1173	1283	1393	1503
400	0.068	3	10	1.062	0.400	1064	1174	1284	1394	1504
400	0.1	1	5	1.125	0.500	1065	1175	1285	1395	1505
400	0.1	1	10	1.125	0.500	1066	1176	1286	1396	1506
400	0.1	3	5	1.062	0.500	1067	1177	1287	1397	1507
400	0.1	3	10	1.062	0.500	1068	1178	1288	1398	1508
400	0.15	1	5	1.125	0.562	1069	1179	1289	1399	1509
400	0.15	1	10	1.125	0.562	1070	1180	1290	1400	1510
400	0.15	3	5	1.062	0.562	1071	1181	1291	1401	1511
400	0.15	3	10	1.062	0.562	1072	1182	1292	1402	1512
400	0.22	1	5	1.375	0.562	1073	1183	1293	1403	1513
400	0.22	1	10	1.375	0.562	1074	1184	1294	1404	1514
400	0.22	3	5	1.312	0.562	1075	1185	1295	1405	1515
400	0.22	3	10	1.312	0.562	1076	1186	1296	1406	1516
400	0.33	1	5	1.375	0.672	1077	1187	1297	1407	1517
400	0.33	1	10	1.375	0.672	1078	1188	1298	1408	1518
400	0.33	3	5	1.312	0.672	1079	1189	1299	1409	1519
400	0.33	3	10	1.312	0.672	1080	1190	1300	1410	1520
400	0.47	1	5	1.625	0.672	1081	1191	1301	1411	1521
400	0.47	1	10	1.625	0.672	1082	1192	1302	1412	1522
400	0.47	3	5	1.562	0.672	1083	1193	1303	1413	1523
400	0.47	3	10	1.562	0.672	1084	1194	1304	1414	1524
400	0.68	1	5	1.625	0.750	1085	1195	1305	1415	1525
400	0.68	1	10	1.625	0.750	1086	1196	1306	1416	1526
400	0.68	3	5	1.562	0.750	1087	1197	1307	1417	1527
400	0.68	3	10	1.562	0.750	1088	1198	1308	1418	1528
400	1.0	1	5	1.875	0.750	1089	1199	1309	1419	1529
400	1.0	1	10	1.875	0.750	1090	1200	1310	1420	1530
400	1.0	3	5	1.812	0.750	1091	1201	1311	1421	1531
400	1.0	3	10	1.812	0.750	1092	1202	1312	1422	1532
400	1.5	1	5	1.625	1.000	1093	1203	1313	1423	1533
400	1.5	1	10	1.625	1.000	1094	1204	1314	1424	1534
400	1.5	3	5	1.562	1.000	1095	1205	1315	1425	1535
400	1.5	3	10	1.562	1.000	1096	1206	1316	1426	1536
400	2.2	1	5	2.125	1.000	1097	1207	1317	1427	1537
400	2.2	1	10	2.125	1.000	1098	1208	1318	1428	1538
400	2.2	3	5	2.062	1.000	1099	1209	1319	1429	1539
400	2.2	3	10	2.062	1.000	1100	1210	1320	1430	1540
400	3.3	1	5	2.625	1.000	1101	1211	1321	1431	1541
400	3.3	1	10	2.625	1.000	1102	1212	1322	1432	1542
400	3.3	3	5	2.562	1.000	1103	1213	1323	1433	1543
400	3.3	3	10	2.562	1.000	1104	1214	1324	1434	1544

1/ Dimensions are bare case sizes (see figure).
2/ Metric equivalents see table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:13Z
Check the source to verify that this is the current version before use.

TABLE II. Metric equivalents of decimal inches. 1/

Inches	MM	Inches	MM	Inches	MM
.031	.79	1.000	25.40	1.812	46.02
.062	1.57	1.062	26.97	1.844	46.84
.312	7.92	1.125	28.58	1.875	47.63
.400	10.16	1.312	33.32	2.062	52.37
.500	12.70	1.375	34.93	2.125	53.98
.562	14.27	1.562	39.67	2.562	65.07
.672	17.07	1.625	41.28	2.625	66.68
.750	19.05	1.781	45.24		

1/ Metric equivalents (to the nearest .01 millimeter) are given for general information only, and are based upon 1 inch = 25.4 mm.

TABLE III. Insulation resistance.

	At dc voltage of	
	200V	400V
In megohms:		
At +25° ±3°C (need not exceed) - - - - - - - -	12,000	12,000
At +125°+4, -0°C (need not exceed) - - - - - - -	150	600
In megohms x microfarads (minimum):		
At +25° ±3°C - - - - - - - - - - - - - - - - - -	2,000	2,000
At +125°+4, -0°C - - - - - - - - - - - - - - - -	10	40

Immersion: Method 104 of MIL-STD-202, condition C.
DWV:
Insulating sleeves - Greater than 4,000 volts, dc.
Terminal to terminal - 200 percent of rated voltage.
Terminals to case (when case is not a terminal) - 200 percent of rated voltage.
IR:
Insulating sleeves - 100 megohms, minimum.
Terminal to terminal - Not less than 50 percent of value specified in table III.
Terminals to case (when case is not a terminal) - 5,000 megohms, minimum.
Cap. - Within ±10 percent of initial value.
DF - Not more than 1.1 percent.

Solderability: Method 208 of MIL-STD-202.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I
50 percent of rated voltage applied.

Moisture resistance: Method 106 of MIL-STD-202.
DWV, IR, Cap., and DF - Same as for immersion.

Terminal strength: Method 211 of MIL-STD-202, conditions A and D.
Condition A - Applied force 5 pounds.
Condition D - 3 rotations of 360 degrees.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:13Z
Check the source to verify that this is the current version before use.

Low temperature and cap. change with temperature:
Low temperature: -55°+0, -3°C for 48 ±4 hours with rated voltage applied.
Cap. change with temperature: At -55°+0, -3°C - -10 percent, maximum. At +125°+4, -0°C - +20 percent, maximum.

Fault count: In accordance with MIL-C-39022.
At 125°+4, -0°C monitored for momentary breakdowns and periods of low IR.
Qualification - 2,000 +72, -0 hours; quality conformance (group C) 240 +48, -0 hours.
Breakdowns - Shall not exceed 6C or 1, whichever is greater (where C is the nominal capacitance in µF of the sample). Total counts for any single capacitor in a 24-hour period shall not exceed 4C or 1, whichever is greater.
Period of low IR - Not to exceed 2 minutes.
IR:
Terminal to terminal - Not less than 33.3 percent of initial requirement.
Terminals to case (when case is not a terminal) - 5,000 megohms, minimum.
Cap. - Within ±10 percent of initial measured value.

Fungus: Method 508 of MIL-STD-810, procedure I.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B (260 ±5°C for 10 ±1 seconds).
IR - See table III.
Cap. - Within ±5 percent of initial value.
DF - Not more than 1.0 percent.

Life: Method 108 of MIL-STD-202.
Qualification:
Accelerated conditions: 140 percent of rated voltage for 2,000 +72, -0 hours.
Rated conditions: 100 percent of rated voltage for 10,000 +96, -0 hours.
DF (at +125°+4, -0°C) between 24 and 48 hours of test - 2 percent, maximum.
During last 48 hours of test - 2.5 percent, maximum.
IR:
Insulating sleeves - 100 megohms, minimum.
Terminal to terminal - Not less than 33.3 percent of initial requirement.
Terminals to case (when case is not a terminal) - 5,000 megohms, minimum.
Cap. - Within ±10 percent of initial value.
DF (at +25° ±3°C) after life - 1.25 percent.

Quality conformance: 240 hours at accelerated conditions and meet the requirements for qualification test.

Extended life:
Accelerated conditions: 2,000 +72, -0 hours.
Rated conditions: 10,000 +96, -0 hours.

Superseded part numbers: The capacitors described herein are considered to be unilaterally interchangeable with those capacitors of the previous issue of this specification sheet.

Marking: In accordance with MIL-C-39022.

Part number: M39022/02- (dash number from table I).

© Cross reference data from type designations of MIL-C-39022/2B to part numbers in MIL-C-39022/2C are not furnished on parts that did not have a qualified source at the time of this revision. (See table IV.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:13Z
Check the source to verify that this is the current version before use.

TABLE IV. Cross-reference data 1/

From Type designation MIL-C-39022/2B	To Part number MIL-C-39022/2C	From Type designation MIL-C-39022/2B	To Part number MIL-C-39022/2C
CHR19A1NE473JL	1057	CHR19A1NE474JL	1081
CHR19A1NE473KL	1058	CHR19A1NE474KL	1082
CHR19A3NE473JL	1059	CHR19A3NE474JL	1083
CHR19A3NE473KL	1060	CHR19A3NE474KL	1084
CHR19A1NE683JL	1061	CHR19A1NE684JL	1085
CHR19A1NE683KL	1062	CHR19A1NE684KL	1086
CHR19A3NE683JL	1063	CHR19A3NE684JL	1087
CHR19A3NE683KL	1064	CHR19A3NE684KL	1088
CHR19A1NE104JL	1065	CHR19A1NE105JL	1089
CHR19A1NE104KL	1066	CHR19A1NE105KL	1090
CHR19A3NE104JL	1067	CHR19A3NE105JL	1091
CHR19A3NE104KL	1068	CHR19A3NE105KL	1092
CHR19A1NE154JL	1069	CHR19A1NE155JL	1093
CHR19A1NE154KL	1070	CHR19A1NE155KL	1094
CHR19A3NE154JL	1071	CHR19A3NE155JL	1095
CHR19A3NE154KL	1072	CHR19A3NE155KL	1096
CHR19A1NE224JL	1073	CHR19A1NE225JL	1097
CHR19A1NE224KL	1074	CHR19A1NE225KL	1098
CHR19A3NE224JL	1075	CHR19A3NE225JL	1099
CHR19A3NE224KL	1076	CHR19A3NE225KL	1100
CHR19A1NE334JL	1077	CHR19A1NE335JL	1101
CHR19A1NE334KL	1078	CHR19A1NE335KL	1102
CHR19A3NE334JL	1079	CHR19A3NE335JL	1103
CHR19A3NE334KL	1080	CHR19A3NE335KL	1104

1/ At the time of this revision only the parts listed on this table had a qualified source.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL
Navy -
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - AS, MC, OS, SH, CG
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5910-1005-2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:13Z
Check the source to verify that this is the current version before use.